

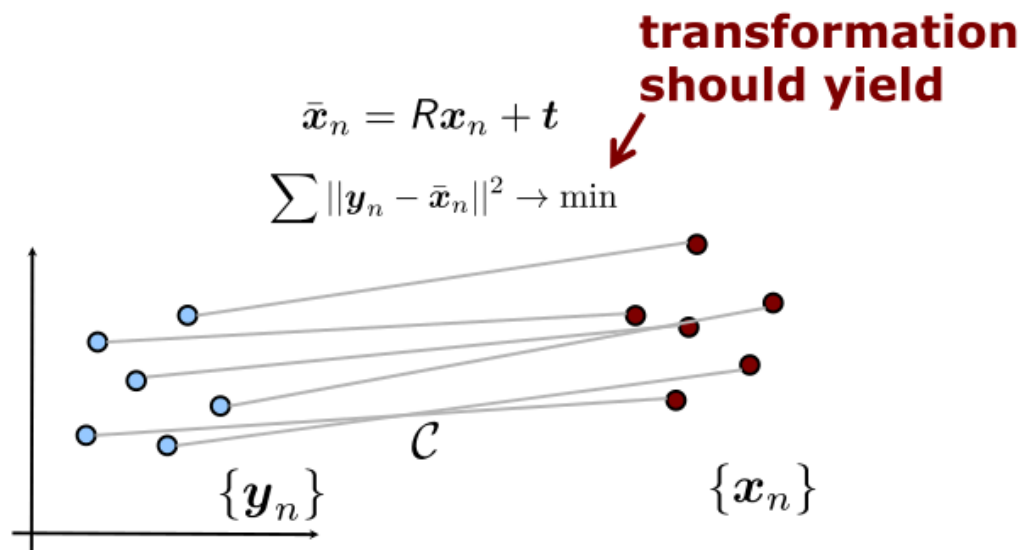
L13/14/15 ICP

Point Cloud Registration & ICP

#1. Known Data Associations

- Find local transformation to best align data points: R and t

Simple Form of Point Cloud Registration

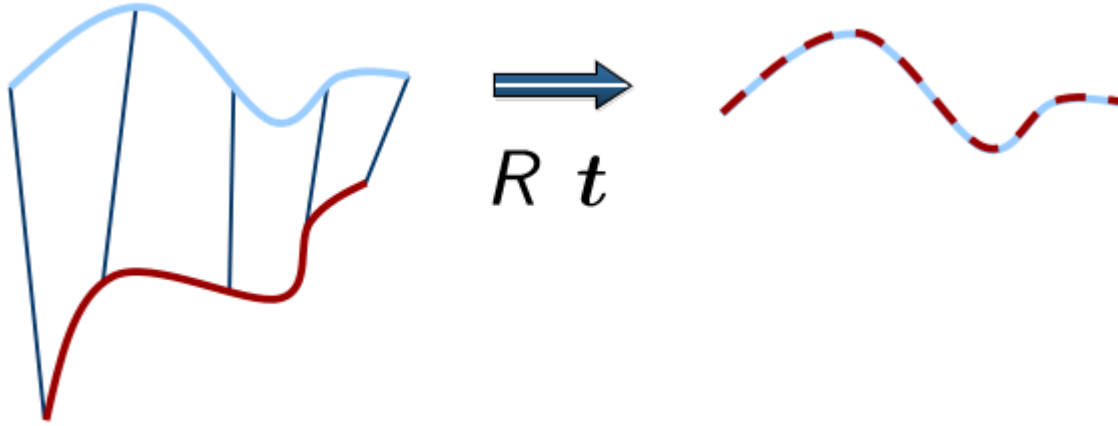


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- Optimisation / Search for Parameters
 - Iterative Closest Point (ICP w/ SVD)
 - Robust Least Squares Approach (#3)
- Known (#1) vs. estimated (#2) correspondences

Basic Alignment Problem

- Given
 - 2 input point sets:
 - $Y = \{y_1, \dots, y_I\}$
 - $X = \{x_1, \dots, x_J\}$
 - Correspondences
 - $C = \{(i, j)\}$
 - Simplify Correspondences: Reorder Point clouds X, Y from C using an index n
- Wanted: Translation t and Rotation R
 - minimising sum of squared errors

$$\circ \sum_{(i,j) \in C} ||\mathbf{y}_i - R\mathbf{x}_j - \mathbf{t}||^2 \rightarrow \min$$



SPECIAL Case of Absolute Orientation Problem

- Absolute Orientation Problem:

$$\bar{\mathbf{x}}_n = \lambda R \mathbf{x}_n + \mathbf{t}$$

- Special case: $\lambda = 1$ Rigid Body transformation:

$$\bar{\mathbf{x}}_n = R \mathbf{x}_n + \mathbf{t}$$

Formal Problem Definition

- Given corresponding points
 - $\mathbf{y}_n, \mathbf{x}_n \quad n = 1, \dots, N$
- and optionally weights:
 - $p_n \quad n = 1, \dots, N$
- Find the parameters $R, \mathbf{t}$ of the rigid body transform with
 - $\bar{\mathbf{x}}_n = R \mathbf{x}_n + \mathbf{t} \quad n = 1, \dots, N$
- so that the squared error is minimised
 - $\sum ||\mathbf{y}_n - \bar{\mathbf{x}}_n||^2 \rightarrow \min$

Direct Optimal Solution exists

- Solving the squared error
- **Direct:** No initial guess needed
- **Optimal:** No better solution exists
- Informal procedure:
 - Compute **shift** from **center of masses**
 - Perform **rotational** alignment using singular value decomposition (**SVD**)

Direct Computation

- Compute means of the point sets:

$$\mathbf{x}_0 = \frac{\sum \mathbf{x}_n p_n}{\sum p_n} \quad \mathbf{y}_0 = \frac{\sum \mathbf{y}_n p_n}{\sum p_n}$$

Rotation Matrix

- Compute cross Covariance matrix based on mean-reduced coordinated

$$H = \sum (\mathbf{x}_n - \mathbf{x}_0)(\mathbf{y}_n - \mathbf{y}_0)^T p_n$$

- Compose H using SVD

$$svd(H) = U D V^T$$

- Compute rotation

$$R = V U^T$$

Translation Vector

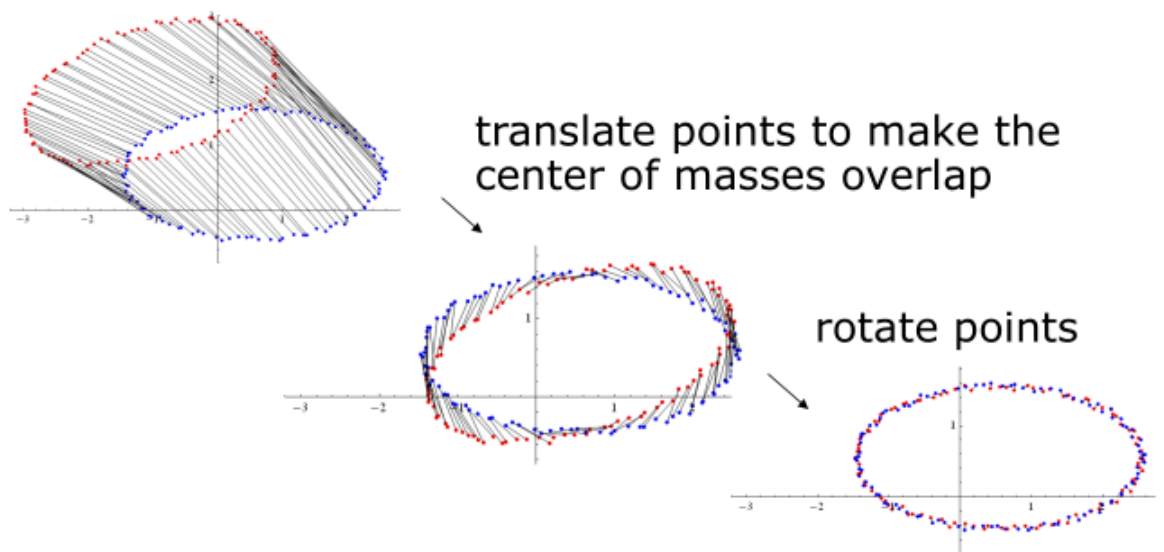
$$\mathbf{t} = \mathbf{y}_0 - R \mathbf{x}_0$$

Note: $\exists$ 2 variants of formulation: Changing U and V $\rightarrow$ changing terms of Cross-Covariance H

SVD-Based Alignment Summary

Alignment through translation and rotation

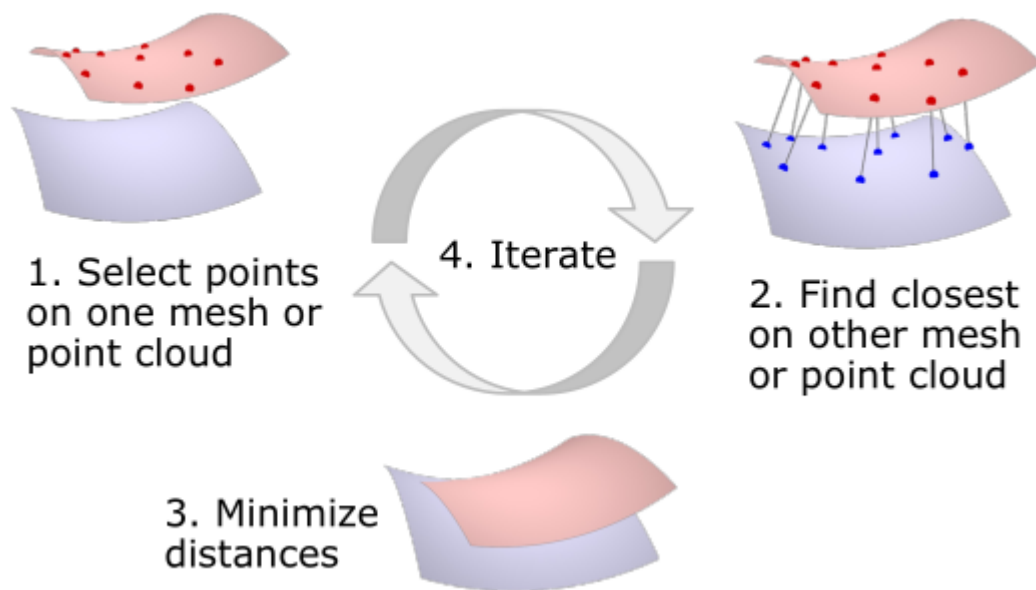
$$\bar{\mathbf{x}}_n = R(\mathbf{x}_n - \mathbf{x}_0) + \mathbf{y}_0$$



Derivation is available in the lecture slides

#2 Point Cloud Registration with Unknown Data Association

- Direct AND Optimal Solution does not exist
- $\rightarrow$ Iterative approach



$$\bar{\mathbf{x}}_n = \mathbf{x}_n$$

$$\text{error } e = \infty$$

while (e has decreased and $e > \text{threshold}$)

$\mathcal{C} = \text{determine_correspondences}(\{\mathbf{y}_n\}, \{\bar{\mathbf{x}}_n\})$

$(\mathbf{t}, R) = \text{compute_transformation_params}(\mathcal{C}, \{\mathbf{y}_n\}, \{\bar{\mathbf{x}}_n\})$

$$\bar{\mathbf{x}}_n = R(\bar{\mathbf{x}}_n - \bar{\mathbf{x}}_0) + \mathbf{y}_0$$

$$e = E(\mathbf{t}, R) = \Phi(R^\top \mathbf{y}_0 - R^\top \mathbf{t}, R)$$

return $\{\bar{\mathbf{x}}_n\}$

ICP Variants

1. Consider Point Subsets
2. Different data association strategies
3. Weight the correspondences
4. Reject potential outlier point pairs

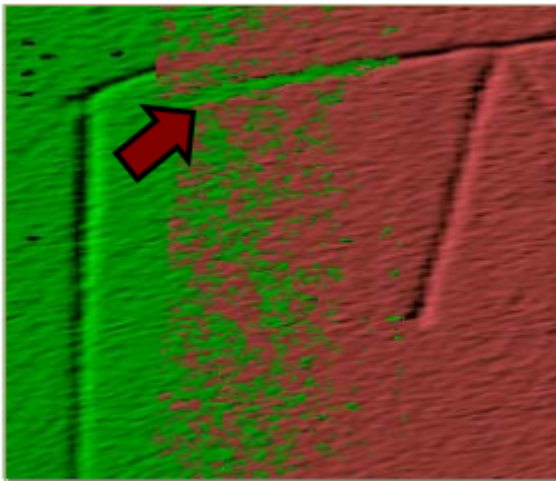
Performance of Variants

- Speed
 - Stability
 - Tolerance w.r.t. noise and outliers
 - Basin of convergence (maximum initial misalignment)
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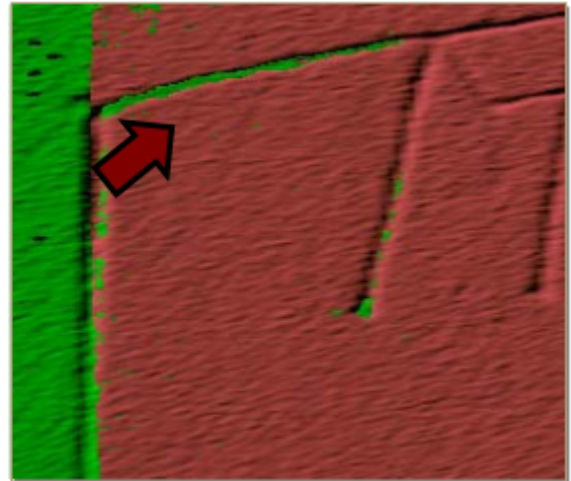
1. Consideration of Point Subsets

- All points

- Random sampling
- Normal-space sampling
 - Samples: Normals distributed as uniformly ap.
 - Good for mostly smooth areas with sparse features

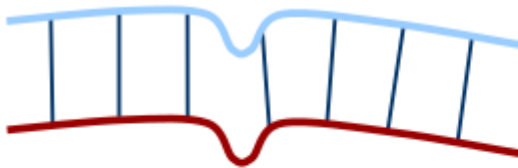


random sampling

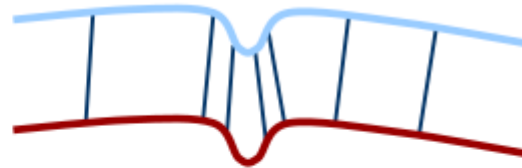


normal-space sampling

- Uniform sub-sampling



uniform sampling



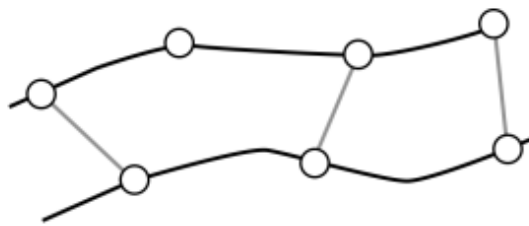
normal-space sampling

- Feature-based sampling
 - Work with only highly distinct points
 - Simplified correspondence search
 - Improved efficiency; (*sometimes*) better accuracy
 - Requires preprocessing

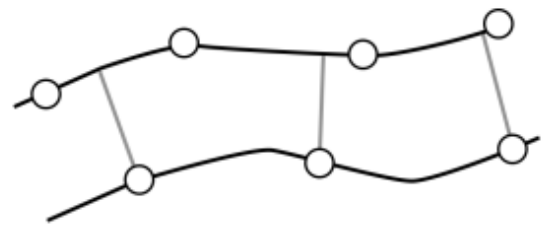
2. Different data association strategies

- Huge impact on convergence and speed
- Investing is key for good results
- Exploit any initial guess
- **Normal-based** metrics **often better** than standard point-to-point metric
- Various different matching methods:
 - Closest point

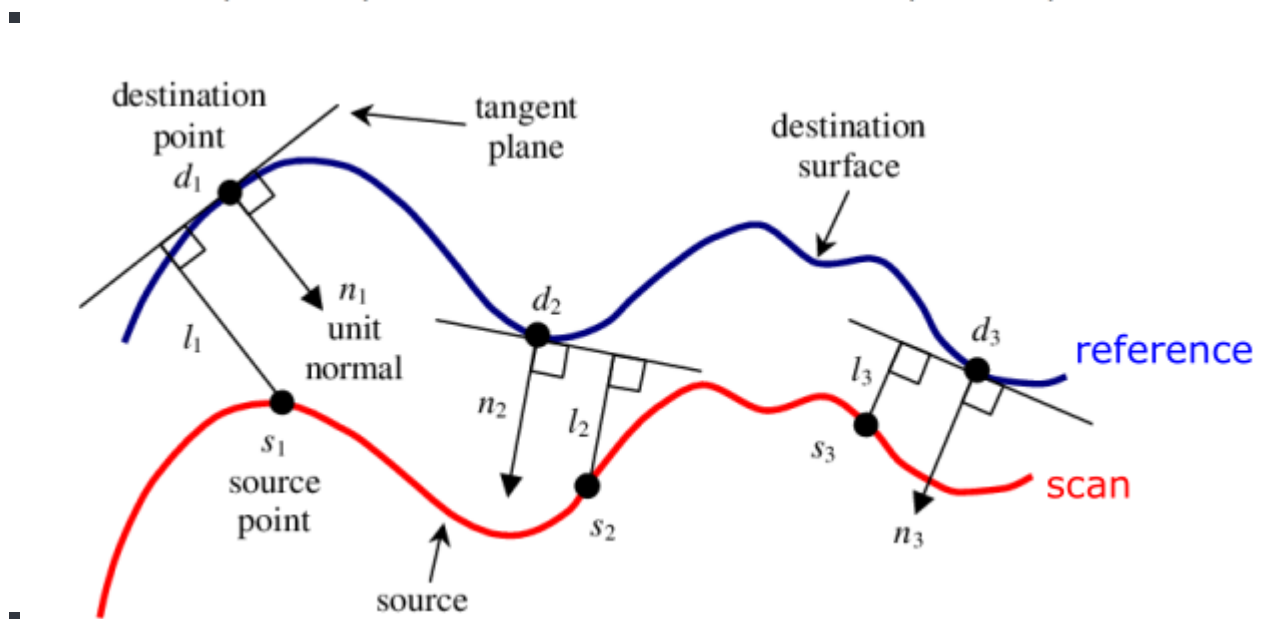
- Using of kd-trees (closest point in other point set)
- Stable, but slow convergence (Vanilla ICP)
- **Without** Initial Guess:
 - align center of masses of both point sets before correspondences
- Closest compatible point
 - Only matching compatible points
 - **Compatibility**
 - Normals
 - Colors
 - Curvature
 - Higher-order derivatives
 - Other local features
- Normal shooting
- Point-to-Plane
 - project Point-to-Point onto the direction of the normal (shot from the found point)



point-to-point



point-to-plane

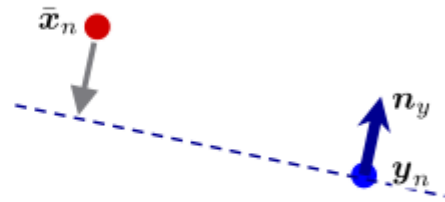


point-to-point



$$\min \sum \| \mathbf{y}_n - \bar{\mathbf{x}}_n \|^2$$

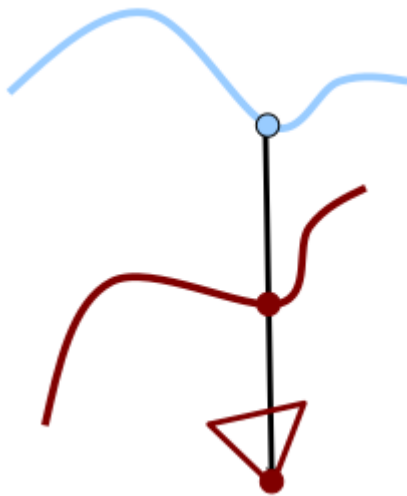
point-to-plane



$$\min \sum ((\mathbf{y}_n - \bar{\mathbf{x}}_n) \cdot \mathbf{n}_y)^2$$

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- Projection-based approaches

- Correspondences by projecting a point from the reference scan towards sensor viewpoint



- etc.

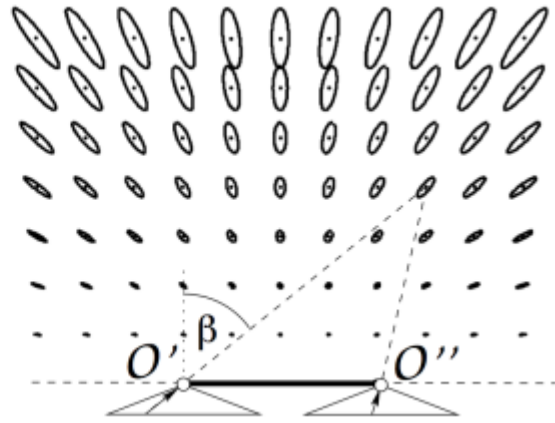
3. Weight the Correspondences

- **Noise:** Weighting based on sensor uncertainty
- Easy to incorporate in the transformation computation

$$\mathbf{y}_0 = \frac{\sum \mathbf{y}_n p_n}{\sum p_n} \quad \mathbf{x}_0 = \frac{\sum \mathbf{x}_n p_n}{\sum p_n}$$

$$H = \sum (\mathbf{x}_n - \mathbf{x}_0)(\mathbf{y}_n - \mathbf{y}_0)^\top p_n$$

- Relevant if measurement noise is varying



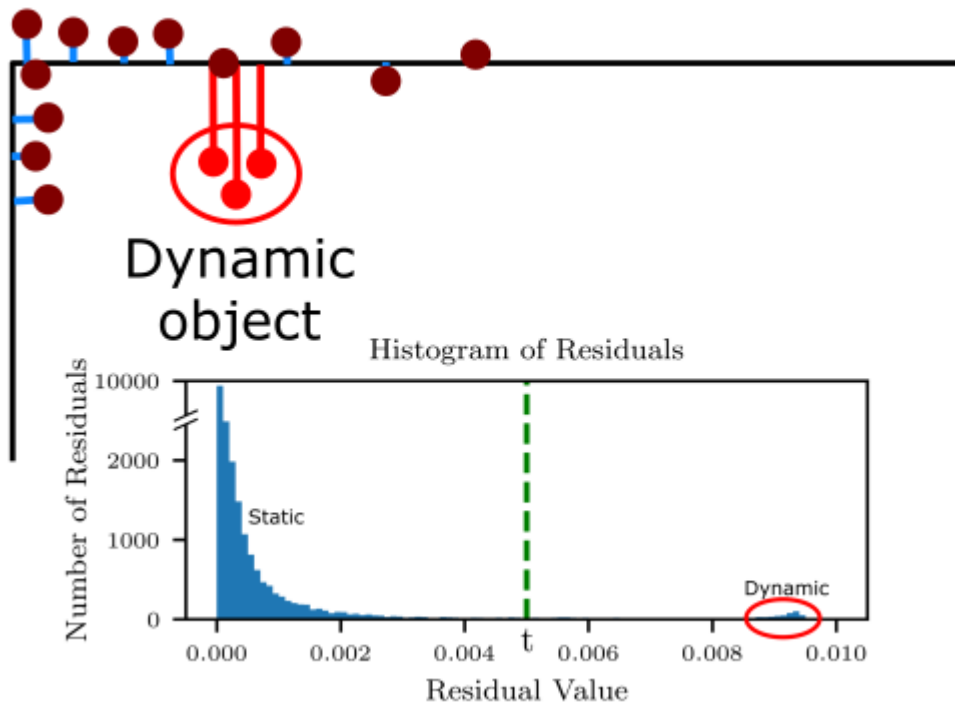
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4. Reject potential outlier point pairs

- P2P distance larger than a given threshold
- Pairs not consistent with neighboring pairs
- Trimmed ICP: Sort correspondences w.r.t. their error, ignore worst t %
 - t related to overlap and outlier ratio
 - Overlap has to be estimated

Mapping in Dynamic Environments

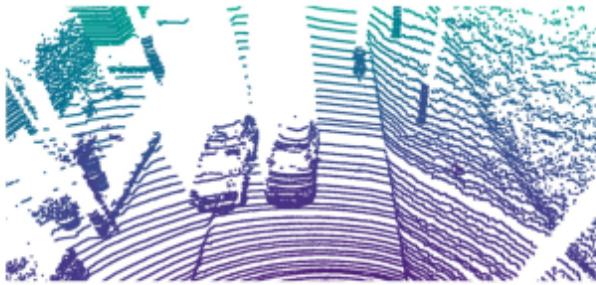
- Detection of Dynamic Elements exploiting residuals



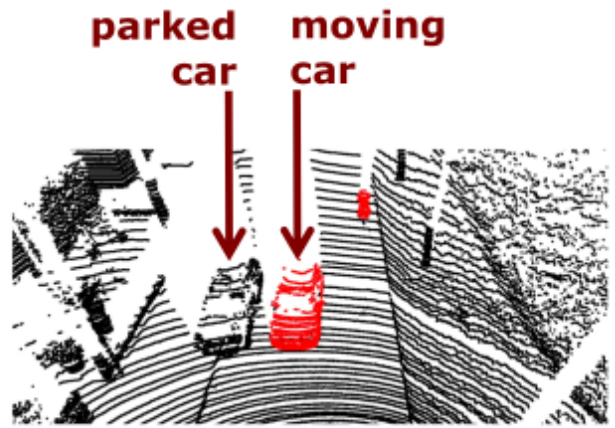
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Learning-based moving Object Detection

- Sophisticated outlier rejection
- Usage for object segmentation in 3D LiDAR scans



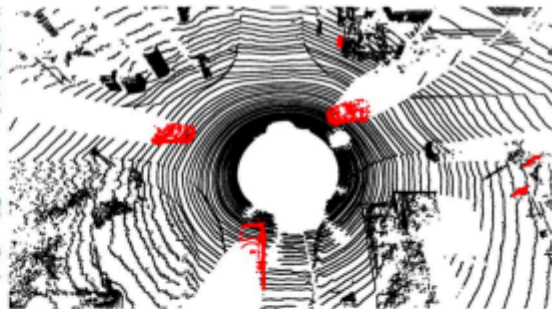
raw point cloud



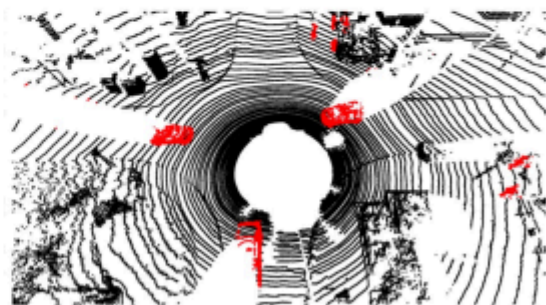
moving object segmentation



raw point cloud



ground truth



estimated moving object segmentation

ICP Algorithm Summary

- Potentiall subsample point clouds
- Determine corresponding points
- Potentially weight or reject outlier pairs
- Compute rotation R , translation t (SVD)
- Apply R and t to all points of the set to be registered
- Compute the error $E(R, t)$
- While error decreased and error > threshold
 - Determine correspondences and weights
 - Compute and apply rigid body transformation
- Output final alignment

- Major problem: Determining the correct data association
- **Often:** Least squares approach with plane-based metric, data association heuristics and outlier rejection

3. Robust Least Squares approach

- SVD assumes Point-to-Point correspondences
- More complex error function -> More general least squares approach
- LS can better consider uncertainties (full 3D point covariances)
- Iterative Gauss Newton-based minimisation

Gauss Newton Minimisation in 2D

- Objective: $\Phi(t_x, t_y, \theta) = \sum ||\bar{\mathbf{x}}_n - \bar{\mathbf{y}}_n|| \rightarrow \min$
- Error Vector: $\mathbf{e}_n = \bar{\mathbf{x}}_n - \mathbf{y}_n = R\mathbf{x}_n + \mathbf{t} - \mathbf{y}_n$
- Parameters: $[t_x, t_y, \theta]^T$
- Linearise the non-linear error function
- Jacobian:
 - $J_n(\mathbf{x}) = \frac{\partial \mathbf{e}_n}{\partial \mathbf{x}}$
 - $J_n = \begin{bmatrix} 1 & 0 & -\sin \theta x_n - \cos \theta y_n \\ 0 & 1 & \cos \theta x_n - \sin \theta y_n \end{bmatrix}$
- Matrix: $H_n = J_n^T J_n$
- Right hand side: $\mathbf{b}_n = J_n^T \mathbf{e}_n$
- Compute terms over all points
 - $H = \sum_n H_n = \sum_n J_n^T J_n$
 - $\mathbf{b} = \sum_n \mathbf{b}_n = \sum_n J_n^T \mathbf{e}_n$
- Solve $H\Delta\mathbf{x} = -\mathbf{b}$ via $\Delta\mathbf{x} = -H^{-1}\mathbf{b}$
- Update parameters $\mathbf{x} \leftarrow \mathbf{x} + \Delta\mathbf{x}$
- Iterative until convergence

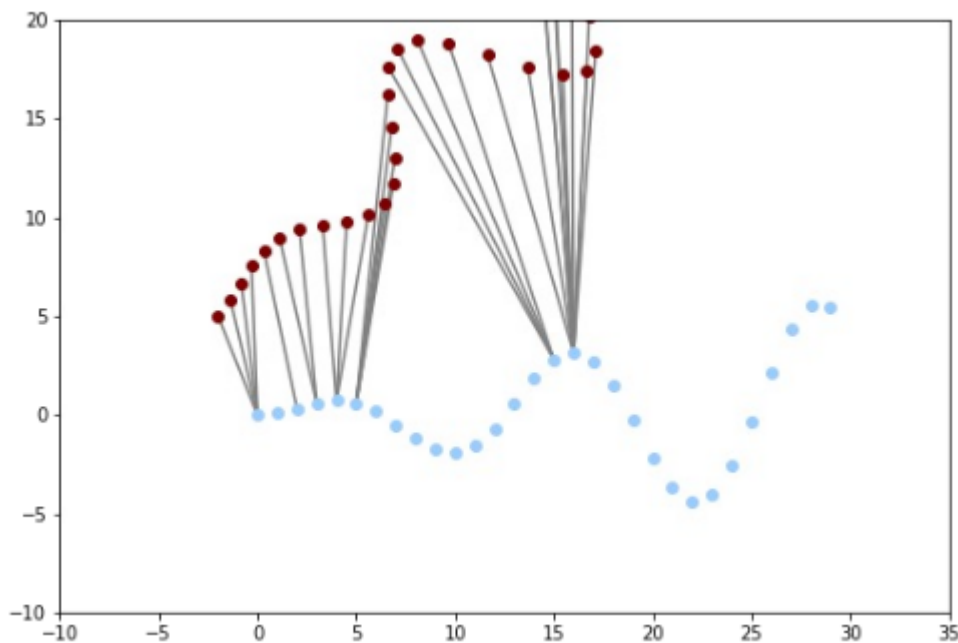
Point-to-Plane LS Registration

$$\begin{aligned} \Phi(t_x, t_y, \theta) &= \sum ||\mathbf{n}_n \cdot \mathbf{e}_n||^2 \\ &= \sum (\mathbf{n}_n \cdot (R(\theta)\mathbf{x}_n + [t_x, t_y]^T - \mathbf{y}_n))^2 \end{aligned}$$

- Different Jacobian
 - Changed objective function leads to a different Jacobian
 - 1D error

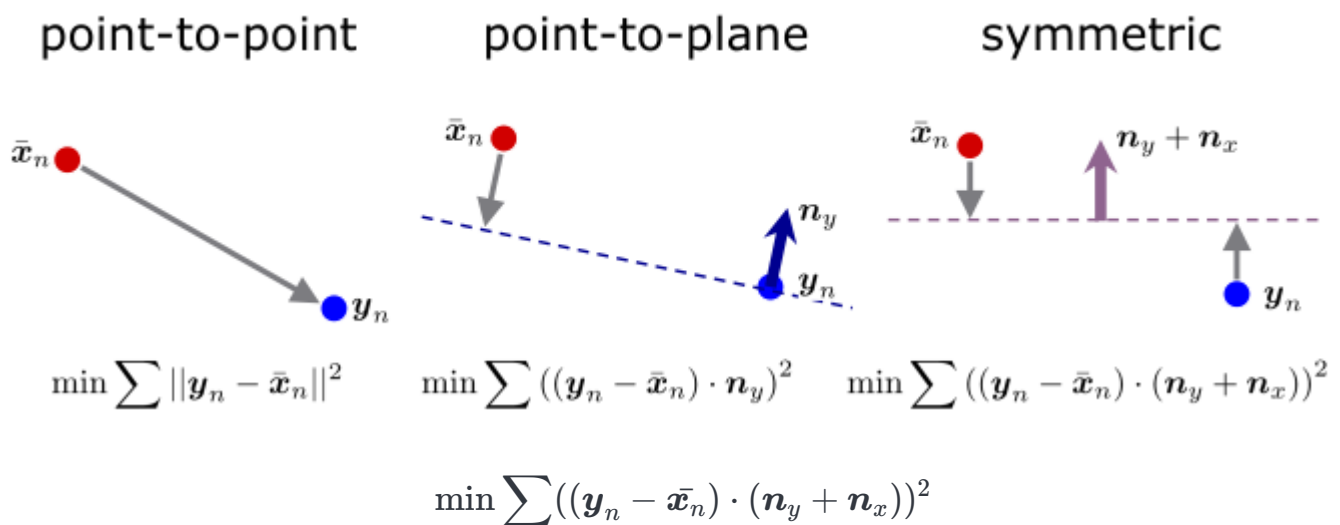
$$J_n(\mathbf{x}) = \left[\frac{\partial e_n}{\partial t_x}, \frac{\partial e_n}{\partial t_y}, \frac{\partial e_n}{\partial \theta} \right]$$

$$= [\mathbf{n}_x, \mathbf{n}_y, \mathbf{n}_x(-t_x \sin(\theta) - t_y \cos(\theta)) + \mathbf{n}_y(t_x \cos(\theta) - t_y \sin(\theta))]$$



Symmetric Point-to-Plane

- Point-to-Plane metric is **NOT** symmetric
- Combination of normals from both surfaces to obtain symmetry

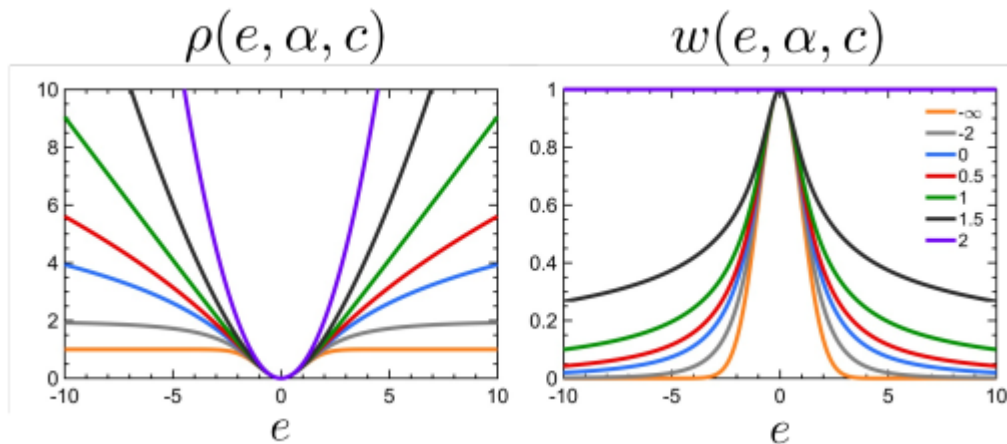


- Additional work required for computing the normals in both clouds (before only in one)
- Leads to an improved performance of ICP

Robust Least Squares

- Data association outliers strongly impact least squares result
- Robust kernels/M-estimators aim at down-weighting the impact of outliers
- Each kernel yields specific weight

- Kernel choice must align with outlier distribution
- Impacts Jacobians



- parameters can be adjusted
- Different Approaches win in different situations

Outlier Rejection is key

- Approaches often use heuristics as initial guess for associations
- Possible question for building heuristic
 - $\exists$ well-identifiable points?
 - Potentially moving objects in scene
 - Exploitation of ego-motion estimates

Remarks from Practice

- Always exploit initial guess (odometry, constant velocity, ...)
 - Exploit multiple sensors
- Normal-based metrics often better than standard point-to-point metric
- Symmetric metric often performs well
- Outdoor scenes may have few planes -> P2Po preferred
- Exploit informed outlier rejection if possible/available
- Adaptive kernels adapt to the outlier situation for each scan pair
- Sanity check to detect failures
 - Vehicle constraints
 - Dynamic constraints
 - ...
- SLAM with loop closing and global optimization is needed
- Proper point uncertainties are often tricky to estimate

Non-Rigid Registration

- Non-Rigid Objects: Deformation thereof
- Location-specific transformations
- Object deformations often encoded via an additional cost term
- Leads to least squares methods with more complex cost functions



- Simplified data association via Skeleton Matching

